30 NOV 2025

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DECODING GAME MARKETS

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RESEARCH BACKGROUND

The video game industry has grown into one of the world's most significant entertainment sectors, producing thousands of titles across diverse genres and platforms. Despite this rapid growth, game performance in global markets varies greatly depending on factors such as genre preference, platform popularity, and market trends.

The purpose of this analysis is to understand what drives global sales performance in the gaming industry. By exploring patterns across genres, platforms, and numerical indicators, this study aims to uncover meaningful insights that help explain which types of games tend to perform well and why.

Through Exploratory Data Analysis (EDA) and statistical modeling, this research investigates:

A. How different genres and platforms contribute to global sales.

B. Whether correlations exist among sales metrics across regions.

C. How statistical models can describe or predict sales performance.

D. What patterns or trends can be extracted to understand industry behavior.

E. This background positions the project not merely as data exploration, but as a structured attempt to understand the economic and behavioral dynamics behind game sales globally.

ANALYSIS METHODOLOGY

This analysis was conducted through several main stages. First, I prepared the data by cleaning it, checking for missing values, correcting data types, and splitting the dataset into three parts according to the analysis requirements: df_clean for regression, df_chi for chi-square, and df_anova for two-way ANOVA.Once the data was cleaned, I performed basic EDA to understand the data structure and variable distributions, and ensure the data was ready for statistical testing.

The next stage was statistical inference, which applied three main methods: Multiple Linear Regression to model Global Sales based on sales per region; Chi-square Test to examine the relationship between Genre and Platform; and Two-Way ANOVA to examine the effect of Genre, Platform, and their interaction on Global Sales.

To validate the results of the inferential analysis, I used descriptive statistics through pivot tables and visualizations such as bar charts, heatmaps, and violin plots. This step helped ensure that the statistical findings were consistent with the descriptive patterns seen in the data. Finally, all findings were analyzed to generate key insights and recommendations relevant to the gaming industry.

EDA

The Global Sales Histogram RESULTS ARE CORRECT and REASONABLE

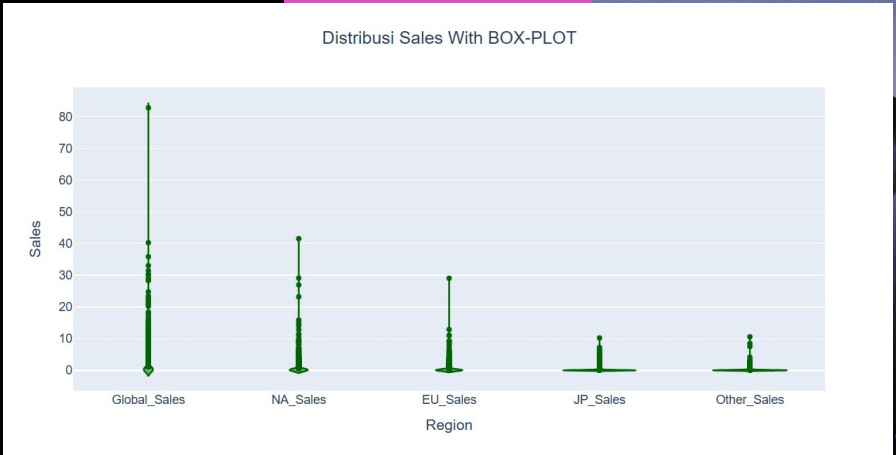
A graph Tell us
(many are 0–2 million, and a few are 80+). That's not a mistake, but the true nature of game sales data:

1. 95% of games don't sell. Only a few games become "super hits" like Wii Sports.
2. Statistically: The distribution is skewed right. This is common for sales data, earnings, viewership, etc.

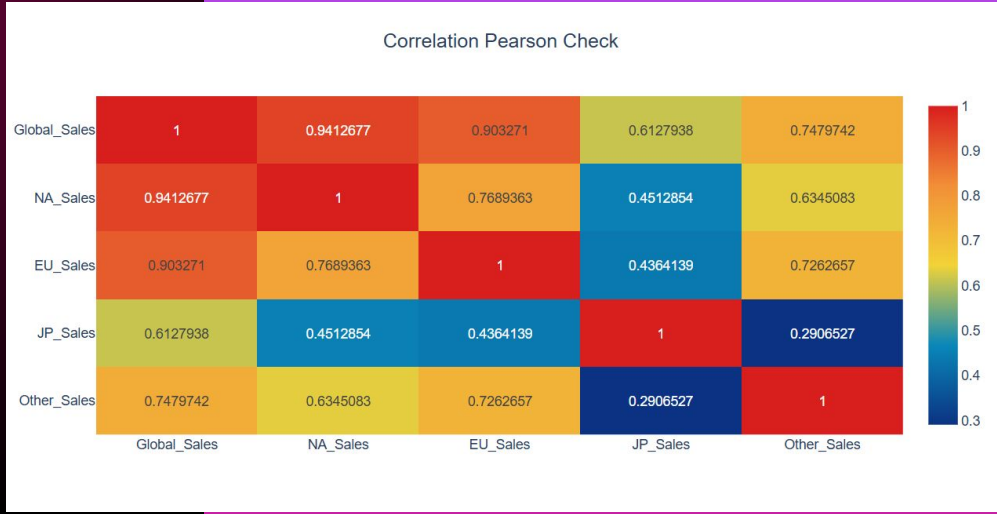
Humanistically:
The majority of games are made, but only a small percentage become huge successes and this is the reality of the industry. Conclusion: The histogram is VALID.

The distribution of sales across all regions, including Global Sales, shows a highly right-skewed pattern.

Most games have low sales (0–2 million copies), while only a few have very high sales (up to 80 million). Outliers are clearly visible, but this is understandable in business terms, given the popularity of some games. Thus, the data is not normally distributed and is predominantly influenced by extreme values.



PEARSON CORRELATION



"Correlation analysis shows that global sales are primarily influenced by North America and Europe, as these two regions have a very strong correlation with total sales. Meanwhile, sales in Japan tend to have a different pattern and do not always follow global trends. This finding confirms that the primary video game markets are in the North and the EU, while Japan operates with more specific preferences."

DIAGNOSTIC MODEL MULTIPLE LINEAR REGRESSION

	coef	std err	z	P>|z|	[0.025	0.975]
const	0.0003	4.37e-05	7.326	0.000	0.000	0.000
NA_Sales	1.0000	7.61e-05	1.31e+04	0.000	1.000	1.000
EU_Sales	1.0000	0.000	4973.517	0.000	1.000	1.000
JP_Sales	0.9999	0.000	5706.917	0.000	1.000	1.000
Other_Sales	0.9996	0.001	1693.361	0.000	0.998	1.001

Omnibus:	214.664	Durbin-Watson:	1.611
Prob(Omnibus):	0.000	Jarque-Bera (JB):	387.159
Skew:	0.047	Prob(JB):	8.50e-85
Kurtosis:	3.748	Cond. No.	9.21

Notes:
[1] Standard Errors are heteroscedasticity robust (HC3)

R-squared:	1.000
Adj. R-squared:	1.000
F-statistic:	2.339e+08
Prob (F-statistic):	0.00
Log-Likelihood:	62632.
AIC:	-1.253e+05
BIC:	-1.252e+05

The linear regression model constructed using the Ordinary Least Squares (OLS) method with robust standard errors type HC3 yielded statistically very strong findings. Based on the estimation results, all predictor variables such as NA_Sales, EU_Sales, JP_Sales, and Other_Sales had a p-value of 0.000, which is much lower than the general significance level ($\alpha = 0.05$). This very small p-value indicates that each regional variable significantly contributes to explaining variation in Global_Sales. In other words, each change in sales in each region has a significant impact on total global sales.

The coefficients for all variables hover around 1.000, which is mathematically consistent because Global_Sales is the direct sum of the four regional sales variables. This explains why the R-squared value reaches 1.000, indicating that the model is able to explain 100% of the variation in Global_Sales without any predictive error.

THE USE OF ROBUST STANDARD ERRORS OF THE HC3 TYPE

Ensures that estimates are not affected by the potential for heteroscedasticity, which often occurs in highly skewed sales data. Despite the uneven distribution of the data, the adjusted standard errors ensure that statistical significance remains valid and is neither overestimated nor underestimated.

Overall, this model is robust, statistically valid, and theoretically produces perfect coefficients due to its mathematical identity data structure. This finding also validates that global sales are indeed formed from the direct contributions of each sales region.

THE VARIANCE INFLATION FACTOR (VIF) CALCULATION RESULTS

	Variable	vif_df
0	const	1.123011
1	NA_Sales	2.640667
2	EU_Sales	3.289803
3	JP_Sales	1.295681
4	Other_Sales	2.195675

Show that all independent variables have VIF values below 5. This indicates that there are no multicollinearity issues in the model. In other words, the regional sales variables (NA, EU, JP, and Other) do not influence each other excessively, so their respective contributions to Global_Sales can be measured stably and accurately. A VIF value in the range of 1–3 indicates that the relationship between regions is reasonable and does not damage the regression model. This model is statistically considered sound, stable, and suitable for further analysis.

Interpretation of the Breusch–Pagan Test (BP Test)

The Breusch Pagan test results show a statistical significance of 115.01 with a *p-value* of 6.18×10^{-24}, which is significantly less than the significance threshold of 0.05. This value indicates that the null hypothesis (H_0) which states that the error variance is constant—is rejected. Thus, the regression model experiences **heteroscedasticity.**

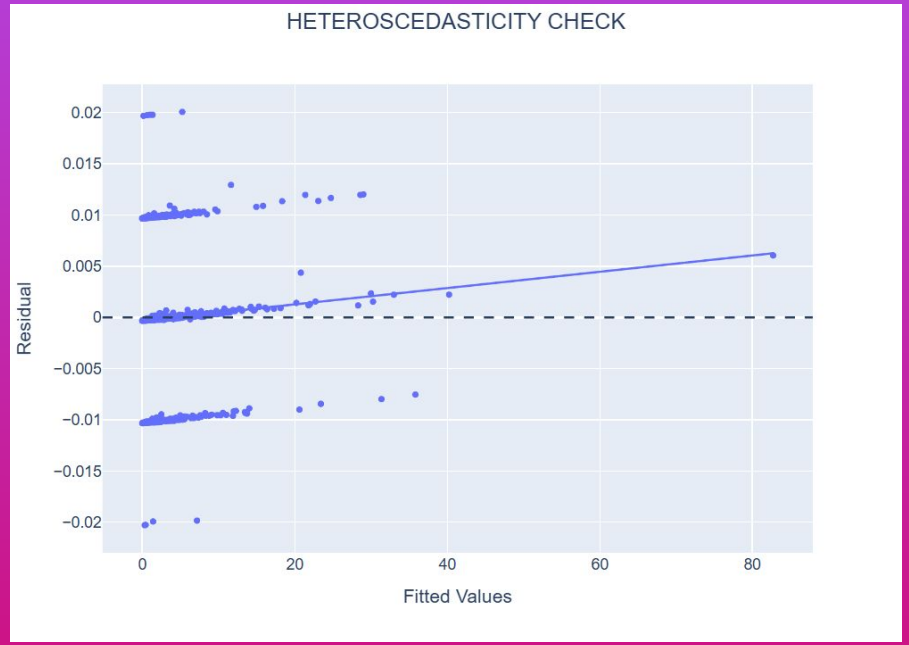
Logically, the presence of heteroscedasticity can be explained by the characteristics of game sales data, which tend to be skewed and have large variations in sales scale across regions. This condition causes the error variance to be non-uniform across all observations. Therefore, using robust standard errors (HC3) in regression estimation is appropriate, as this method is capable of producing more stable and unbiased standard errors even when the homoscedasticity assumption is not met.

The Plot of Residuals Against Fitted Values

That Graph shows a non-random pattern, where the residual distribution appears to form several horizontal lines that then widen as the fitted value increases. Furthermore, the lowess trend line appears to slope upward, indicating that the residual variance tends to increase at larger fitted values.

This pattern indicates that the error variance is not constant across the model's prediction range. In other words, the regression model visually exhibits heteroscedasticity, consistent with the Breusch–Pagan test results, which also reject the assumption of homoscedasticity.

This heteroscedasticity condition is common in game sales data, which is highly skewed and has varying sales scales across regions. Therefore, regression estimation using robust standard error HC3 remains an appropriate approach to ensure valid statistical inferences.



CONCLUSION OF THE MEANING OF THE TEST WITH THE EVIDENCE FROM THE PICTURE

```
RMSE: 0.005229827679224132
R2 Score: 0.9999888425084879
MAPE: 0.02497315162735957
```

Based on the results of the linear regression model evaluation. The RMSE value of 0.00523 indicates that the average model prediction error for Global Sales is very small. The R^2 value of 0.99999 indicates that almost all of the variation in Global Sales can be explained by sales from each region (NA, EU, JP, and Other). Furthermore, the MAPE value of 2.5% confirms that the model has a very high level of prediction accuracy. Overall, the model can be considered very good at predicting Global Sales, and the use of robust standard errors (HC3) ensures that statistical inferences remain valid despite indications of heteroscedasticity.

CONCLUSION FROM CHI SQUARE TEST

```
Chi-square: 5609.930850083323
p-value: 0.0
Degrees of freedom: 253
```

The chi-square test results showed a very large chi-square statistic value with a p-value < 0.05. This indicates a significant relationship between platform and game genre.In other words, game genres are not evenly distributed across all platforms. Certain platforms have strong biases toward certain genres. For example, some platforms are more dominant for action or sports games, while others are more focused on role-playing or adventure games. These findings suggest that platform influences the types of games released and their popularity, so these two variables need to be analyzed together when understanding game industry trends and developer behavior.

CONCLUSION FROM TWO WAY ANOVA TEST

Based on a two-way ANOVA test on four genres and four platforms with Global Sales as the dependent variable, the following results were obtained:

1. Genre does not have a significant effect on Global Sales (p = 0.352, np2 = 0.0008). This means that differences in game genres are not the main factor determining global sales.

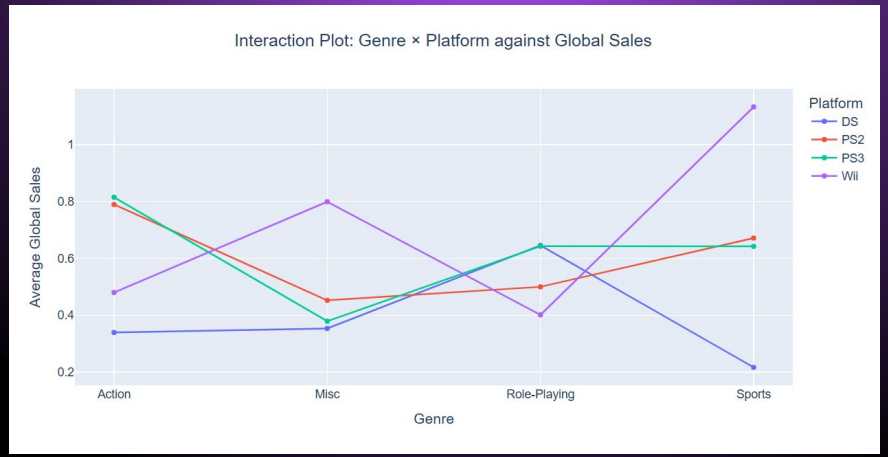
2. Platform has a significant effect on Global Sales (p = 0.000283), but the effect is small (np2 = 0.005). This indicates that some platforms do have higher average sales, but the difference is not significant in practical terms.

3. There is a significant interaction between Genre and Platform (p = 0.0029, np2 = 0.0065), but the effect is also small. This means that sales patterns between genres can vary depending on the platform, although not significantly.

Overall, the Platform and Genre–Platform interactions are statistically significant, but the strength of both effects is small, so the explanation of variance in Global Sales by these two factors is relatively limited.

	Source	SS	DF	MS	F	p-unc	np2
0	Genre	13.044980	3.0	4.348327	1.088344	0.352676	0.000858
1	Platform	75.795300	3.0	25.265100	6.323608	0.000283	0.004965
2	Genre * Platform	100.340591	9.0	11.148955	2.790474	0.002911	0.006562
3	Residual	15190.364748	3802.0	3.995362	NaN	NaN	NaN

THE TWO-WAY ANOVA RESULTS

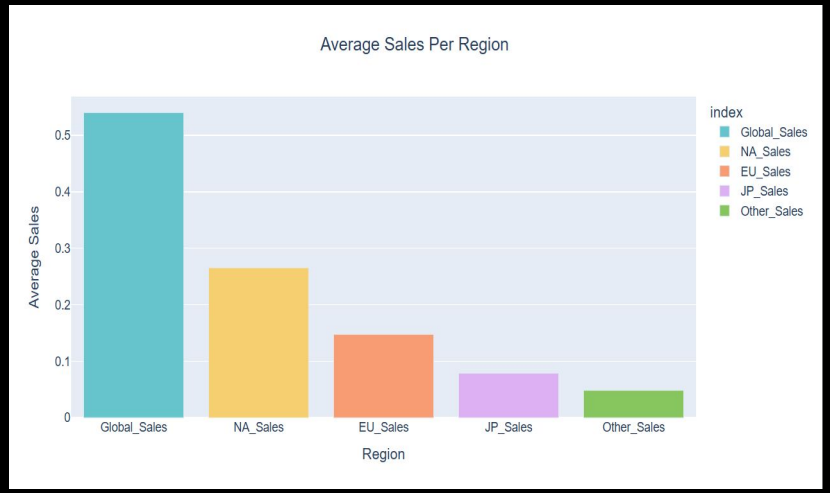


Show that Platform has a significant effect on Global Sales ($p < 0.001$), while Genre has no significant effect ($p = 0.35$). There is a significant interaction between Genre and Platform ($p = 0.002$), which means that the impact of Genre on sales varies depending on the Platform used. However, the overall effect size (partial η^2) for all factors is relatively small, so the strength of the influence is limited. The interaction plot shows a pattern of differences in average Global Sales between Genres that changes according to Platform, supporting the results of the interaction test.

STATISTICS DESCRIPTIVE OVERVIEW

INSIGHT AVERAGE SALES PER REGION

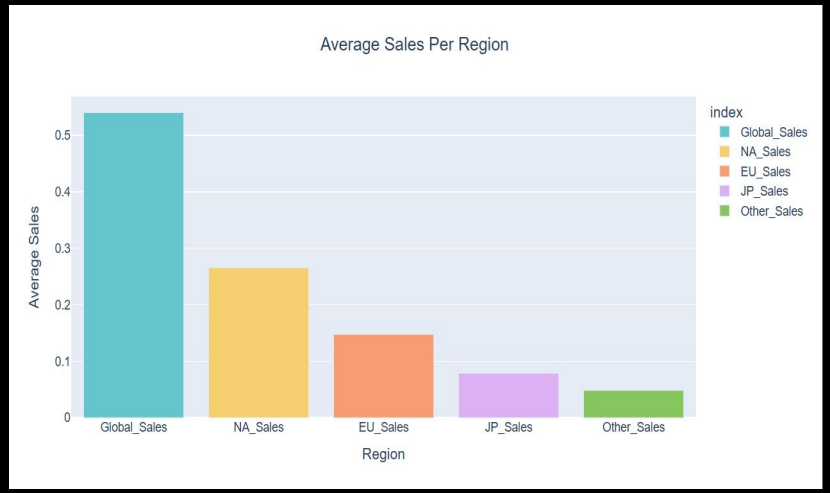
Analysis of average sales per region reveals a highly unbalanced market structure. Overall global sales achieve the highest average value due to the accumulation of all regions, but the market that truly drives the industry is North America, followed by Europe. These two regions consistently contribute the largest to total sales. Conversely, Japan and Other Regions have significantly lower average sales, reflecting their smaller market scale and more specific preferences.



This pattern is consistent with correlation and regression findings: NA and EU have the strongest relationship with global sales, while JP has a more unique pattern and does not fully follow global trends. The highly right-skewed sales distribution also confirms that only a small percentage of games are able to penetrate this key market at scale. Overall, these results help validate the inferential statistics finding that the global sales structure is indeed heavily dependent on Western market dynamics.

RECOMMENDATION AVERAGE SALES PER REGION

Based on the observed sales patterns, the company needs to focus its marketing, distribution, and content development strategies on the NA and EU regions, as these two markets show clearly higher sales potential than other regions. Meanwhile, Japan remains important but requires a separate approach due to its distinct genre preferences and gaming culture. From a business perspective, this strategic division helps the company allocate budgets more efficiently and maximize returns from each region.

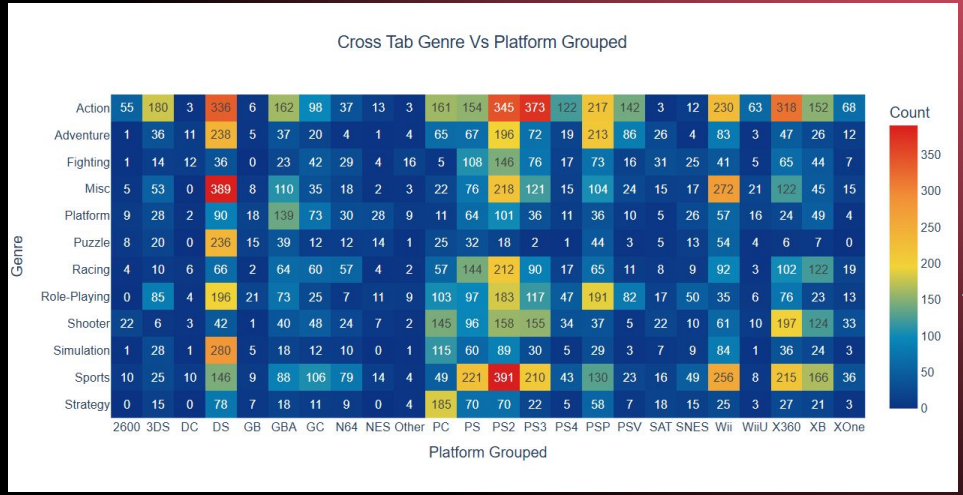


From a socio-cultural perspective, the differences in responses between markets demonstrate that a game's value does not stand alone, but is shaped by the social, cultural, and symbolic contexts within that society. By understanding how each region interprets entertainment products in its own way, the company can design games and campaigns that are not only commercially competitive but also resonate more deeply with players.

INSIGHT Cross Tab Genre Vs Platform Grouped

The visualization shows that some genres, such as Action, Role-Playing, Shooter, and Sports, have very broad and dominant distribution across popular platforms like the PS2, PS3, DS, and X360. These genres tend to be "cross-platform," meaning they are accepted by a wide variety of gamers.

Meanwhile, genres like Puzzle, Adventure, and Strategy have much more limited distribution and tend to be strong only on specific platforms, particularly handheld platforms like the DS. This pattern suggests that the fit between game format and platform characteristics influences game distribution across the industry.

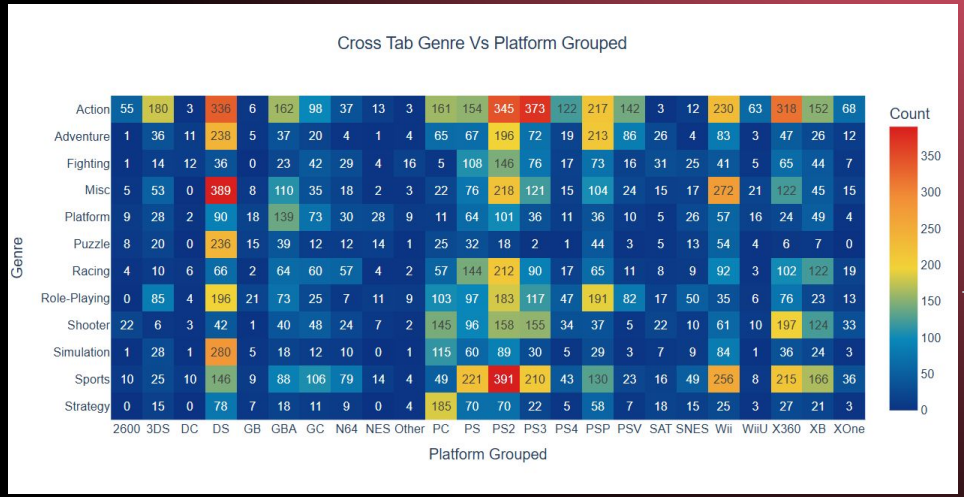


Genre	2600	3DS	DC	DS	GB	GBA	GC	N64	NES	Other	PC	PS	PS2	PS3	PS4	PSP	PSV	SAT	SNES	Wii	WiiU	X360	XB	XOne
Action	55	180	3	336	6	162	98	37	13	3	161	154	345	373	122	217	142	3	12	230	63	318	152	68
Adventure	1	36	11	238	5	37	20	4	1	4	65	67	196	72	19	213	86	26	4	83	3	47	26	12
Fighting	1	14	12	36	0	23	42	29	4	16	5	108	146	76	17	73	16	31	25	41	5	65	44	7
Misc	5	53	0	389	8	110	35	18	2	3	22	76	218	121	15	104	24	15	17	272	21	122	45	15
Platform	9	28	2	90	18	139	73	30	28	9	11	64	101	36	11	36	10	5	26	57	16	24	49	4
Puzzle	8	20	0	236	15	39	12	12	14	1	25	32	18	2	1	44	3	5	13	54	4	6	7	0
Racing	4	10	6	66	2	64	60	57	4	2	57	144	212	90	17	65	11	8	9	92	3	102	122	19
Role-Playing	0	85	4	196	21	73	25	7	11	9	103	97	183	117	47	191	82	17	50	35	6	76	23	13
Shooter	22	6	3	42	1	40	48	24	7	2	145	96	158	155	34	37	5	22	10	61	10	197	124	33
Simulation	1	28	1	280	5	18	12	10	0	1	115	60	89	30	5	29	3	7	9	84	1	36	24	3
Sports	10	25	10	146	9	88	106	79	14	4	49	221	391	210	43	130	23	16	49	256	8	215	166	36
Strategy	0	15	0	78	7	18	11	9	0	4	185	70	70	22	5	58	7	18	15	25	3	27	21	3

Recommendation Cross Tab Genre Vs Platform Grouped

Companies can maximize market potential by prioritizing the development and porting of universal genres like Action and Role-Playing to major platforms with a broad user base, as these genres have proven adaptability and appeal across devices. Meanwhile, more niche genres like Puzzle or Strategy should be directed to platforms that already have a suitable audience, allowing for more efficient use of resources.

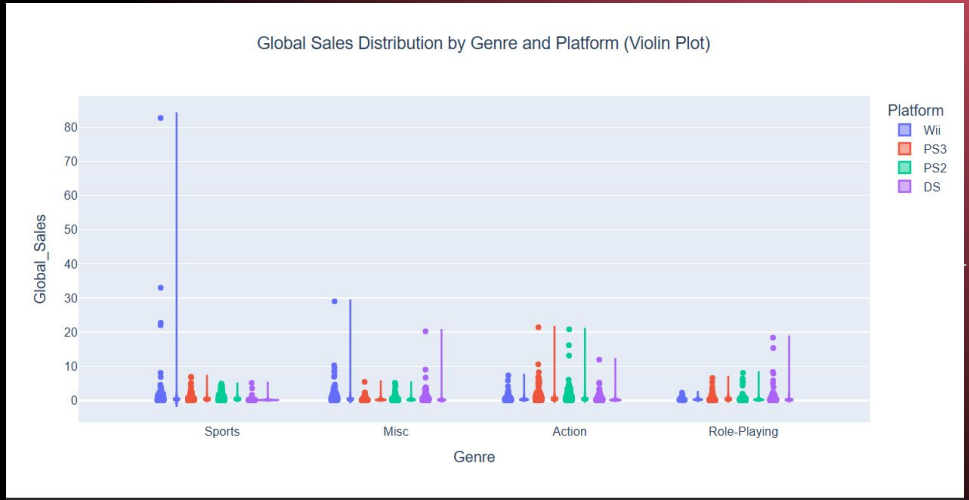
From a philosophical perspective, this distribution pattern reminds us that player preferences are shaped by habitus—their habits, technological context, and the platform ecosystem they inhabit. By understanding that the “meaning of play” differs across platforms, companies can design strategies that better align with the user experience and value they seek, rather than simply following production numbers.



Insight Cross Tab Genre Vs Platform Grouped

Violin plots show distinctly different sales patterns across genres. The Sports genre exhibits strong dominance on the Wii, with significantly higher distribution and a few outliers at extreme sales levels, indicating that this platform has a particular appeal for games that emphasize movement and physical activity.

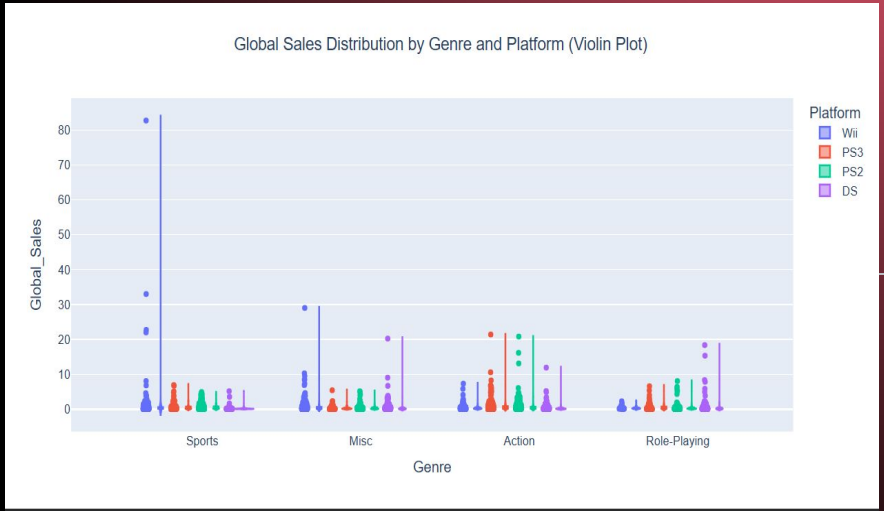
Genres like Action and Role-Playing, on the other hand, have a more even distribution pattern across platforms, with no single platform clearly dominating. The DS stands out for Role-Playing, with one significant outlier, while the Wii is at its lowest for the same genre. These findings indicate that platform characteristics play a significant role in shaping player preferences, and the relationship between genre and medium is not always uniform.



Recommendation Cross Tab Genre Vs Platform Grouped

Companies can maximize their release strategy by targeting specific genres to platforms that best suit their technical characteristics and audience. For sports, the Wii has proven to be the most responsive platform, making exclusive releases or a focus on motion features effective. Conversely, genres like role-playing offer significant opportunities on handheld platforms like the DS, where players tend to seek more personalized and mobile narrative experiences.

For genres with widespread distribution like action, a multi-platform strategy is the best approach to expanding market reach. From a philosophical perspective, these results remind us that the gaming experience is always shaped by the interaction between players and the medium they use. Understanding the "platform context"—that is, how technology influences how people experience games—helps companies make decisions that are not simply based on numbers but also aligned with players' values, comfort, and behavioral patterns.



FINAL CONCLUSION AND REFLECTION

This analysis shows that game sales patterns are heavily influenced by the fit between genre and platform characteristics. Genres like Sports perform particularly strongly on motion-based platforms like the Wii, while Role-Playing games are more prominent on handheld platforms like the DS.

Other genres, like Action, exhibit a more even distribution, indicating a broad audience across multiple platforms. Through EDA and statistical analysis, these patterns can be more clearly identified, yielding deeper and more relevant insights. Overall, a better understanding of the relationship between genre, platform, and sales distribution allows companies to create more targeted release strategies, based on accurate data and sound interpretation.

REFLECTION

1. Why is EDA important before advanced statistical analysis?

EDA is crucial because it helps us understand the underlying structure of the data before undertaking more complex analysis. By conducting EDA, we can identify initial patterns, data distributions, anomalies, outliers, and quality issues such as missing values. This initial understanding ensures that the results of statistical analyses such as correlation or regression are not biased or misleading. EDA also helps formulate more precise analytical questions, resulting in more accurate and relevant decisions in subsequent stages.

2. How does the process of selecting datasets, preprocessing, and applying analytical techniques prepare you for the working world?

This process mirrors the data workflow in the professional world: from selecting datasets relevant to business objectives, cleaning the data to make it usable, to applying statistical analysis to answer meaningful questions. In real-world work, data is often messy, so preprocessing skills are crucial to ensuring quality insights. Techniques like correlation, chi-square, or regression help understand relationships between variables and support evidence-based decision-making. For example, in the gaming or e-commerce industries, analysts need to determine which factors most influence sales; without an understanding of EDA and statistical analysis, business decisions can be flawed. Through this process, I learned to work more systematically, critically, and purposefully—core competencies that are essential in the data world.

THANK YOU

DATA SCOURCE :

https://www.kaggle.com/datasets/volodymyrpivoshenko/video-game-sales-dataset

LINK CODE GOOGLE COLAB SCOURCE :

https://colab.research.google.com/drive/1zTZqoUoZ9BE-2ykAQvzuZHmsw1h-FDyV?usp=sharing